

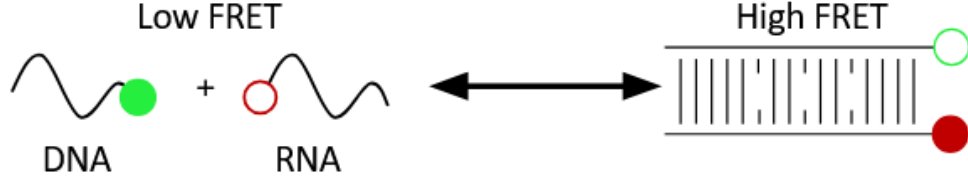
# GCRC Hackathon 1

Martin Jasinski, Nane Vardanyan, Cynthia Shaheen

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## 1 Background

Antisense oligonucleotides (ASOs) are promising therapeutic molecules for targeting disease-associated RNA transcripts, but evaluating their specificity and molecular interactions remains challenging. Conventional bulk assays primarily characterize ASO-RNA interactions at equilibrium, limiting their ability to resolve the dynamic association and dissociation events that govern target engagement. Here, we are developing a single-molecule platform that combines CLiC nanowell confinement with Förster resonance energy transfer (FRET) to characterize ASO-RNA interactions in real time.



## 2 Problem Statement

You are provided with a dataset consisting of single-microscopy images of two complementary RNA strands diffusing in nanowells, shown in the graphic in Fig. 1. The RNAs have fluorophores on the 5' ends of each RNA and are excited with a **647 nm** and **532 nm** laser respectively. The captured images as sequences such that they are labelled with alternating frames in the following pattern:

$$V : \{F_1, F_2, F_3, F_4, F_5 \dots\} \quad (1)$$

where **red** signifies the use of the 647 laser, **green** signifies the use of a 532 laser, and **sienna** signifies use of both lasers.

Your task is simple, analyze the videos and to find the transverse positions  $x, y$  of each particle in each frame, and determine colocalized videos.

The output of the code should consist of a CSV file that consists of the following parameters:

Table 1: Example: Output CSV structure

| Particle | Video | $xy_1(t)$                         | $xy_2(t)$                         | $I_1(t)$                | $I_2(t)$               |
|----------|-------|-----------------------------------|-----------------------------------|-------------------------|------------------------|
| P01      | Vid_1 | [(12.4, 45.2), (12.2, 47.1), ...] | [(12.6, 45.8), (10.2, 43.1), ...] | [1.23, 3.24, 1.5, ...]  | [1.23, 3.22, 1.0, ...] |
| P02      | Vid_1 | [(89.1, 12.3), (50.2, 30.1), ...] | [(89.0, 12.9), (88.2, 13.1), ...] | [3.43, 0.34, 1.21, ...] | [7.47, 4.56, 6.9, ...] |
| P03      | Vid_2 | [(34.7, 78.1), (30.2, 77.1), ...] | [(35.2, 77.9), (33.2, 78.1), ...] | [4.13, 1.55, 1.73, ...] | [2.67, 5.21, 8.4, ...] |
| ⋮        | ⋮     | ⋮                                 | ⋮                                 | ⋮                       | ⋮                      |